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Big Data-Driven Evaluation of Acid Fracturing Effectiveness in Deep Carbonate Reservoirs

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Abstract

The carbonate reservoirs in Penglai gas area have characteristics such as deep burial depth (6000-8000m), poor physical properties (porosity of 2%~6%), and strong heterogeneity. Early-stage acid fracturing results are highly disparate, and the key factors dictating treatment success remain poorly understood.

This study develops a workflow to evaluate deep carbonate rock after acid fracturing by analyzing big data. Following data curation of geological, construction, and production metrics from pre-fracturing wells, feature importance was ranked using Random Forest with the SHAP (Shapley Additive Explanations) model. These rankings were amalgamated via a weighted average to produce a consensus parameter hierarchy, guiding neural network identification of key drivers. A subsequent hybrid AHP-EWM framework was developed for multivariate quantitative evaluation, elucidating the principal factors governing acid fracturing effectiveness and establishing a novel assessment protocol for Penglai gas area carbonate wells.

Results demonstrate that treatment section length, acid intensity, acid volume, total hydrocarbon content, pressure drop frequency, significantly impact fracturing effectiveness. The constructed quantitative acid fracturing effectiveness index achieved over 90% linear correlation with actual well test production, verifying model validity. The acid fracturing effectiveness index ranges from 0.22 to 0.99, the geological index spans 0.15 to 0.55, and the engineering index varies between 0.18 and 0.51. Rational acid fracturing optimization and high-quality field implementation constitute dual critical factors for ensuring optimal stimulation performance in the Penglai Gas Zone.

This study develops a novel, multifaceted evaluation method for assessing acid fracturing outcomes in deep carbonate reservoirs. This approach lays the theoretical groundwork for designing highly efficient deep acid fracturing treatments.

Keywords: Carbonate reservoirs, Big Data Analysis, Random Forest, Analytic Hierarchy Process, Acid fracturing effectiveness

Introduction

The Sichuan Basin is one of the basins with the greatest potential for natural gas exploration and exploitation in China, boasting a total resource volume of $39.94 \times 10^{12} \text{m}^3$. Within this basin, the natural gas resource volume in deep to ultra-deep marine carbonate formations amounts to $14.33 \times 10^{12} \text{m}^3$ ^[1–3]. The Penglai gas area is a paleo uplift in the central part of the Sichuan Basin, where 9 natural gas-producing layers have been discovered. Among them, the geological reserves of natural gas for the Anyue gas field amount to $1.03 \times 10^{12} \text{m}^3$. Acid fracturing technology has emerged as an important measure to boost productivity for such oil and gas reservoirs and sustain their stable production. To enhance the effectiveness of acid fracturing technology, assessing the outcomes post-acid fracturing construction is essential^[4].

Enhancing acid fracture design necessitates an effective model capable of predicting the resulting productivity. Thus, evaluating post-fracturing well performance is critical, and a range of methods have been developed to quantify improvements in well productivity. The most commonly employed methods for estimating well productivity include empirical correlations, numerical reservoir simulation, and integrated models^[5]. However, empirical and semi-empirical correlations often yield substantial errors in conductivity estimation when applied to different rock types, primarily due to variations in fracture patterns. Analytical models, on the other hand, require the introduction of artificial assumptions, such as constant temperature and Newtonian fluid behavior, which deviate significantly from actual downhole conditions. Numerical simulation methods impose high demands on the accuracy of input data, and even minor errors in these inputs can lead to severe inaccuracies in the calculation results. Moreover, there is currently a lack of mature methodologies for the precise identification and characterization of subsurface fracture-cavity reservoirs.

Integrated models have been utilized to conduct performance evaluations and productivity predictions for multi-stage acid fracturing treatments. These models typically take into account hydraulic fracture propagation, wormhole development, complicated fluid leak-off media, and heat transfer behaviors^[6, 7]. Nevertheless, such models are often highly complex and require solutions via analytical, semi-analytical, or numerical methods^[8–10], which are computationally intensive and time-consuming. Furthermore, integrated models are only effective and applicable in specific regions.

Diverse machine learning models have been developed to target carbonate rocks, leveraging large volumes of experimental data to predict acid fracture conductivity and well productivity^[11]. Forecasting acid fracturing performance to support optimal acid fracturing design demands a comprehensive understanding of acid-rock reaction processes, rock strength characteristics, rock stress, and their combined impacts on fracture conductivity. Existing predictive models, however, exhibit significant limitations in describing the conductive capacity of acid-etched fractures under stress-induced closure conditions. In response, artificial intelligence (AI) approaches have been proposed to enhance the prediction precision of acid-fracture conductivity. Specifically, artificial neural networks (ANN) and adaptive network-based fuzzy inference systems (ANFIS) have been utilized in the development of intelligent predictive models that can yield accurate designs for acid-fracture conductivity^[12]. For instance, an ANN-based model trained on over 3,000 scenarios (encompassing reservoir properties and treatment parameters) was proven effective in estimating acid fracturing effectiveness in naturally fractured reservoirs^[13]. The broader applications of AI techniques—including ANN, fuzzy logic (FL), random forest (RF), and support vector machines (SVM)—have been demonstrated in hydraulic fracturing design and optimization^[14, 15]. These techniques enable the optimization of key parameters such as treatment stages, proppant/fluid volumes, chemical additive types, and the identification of sweet spots. A notable limitation, however, is that most AI-developed models overlook critical geological and reservoir properties^[16]. Recent advancements have addressed this issue by integrating complex interactions between natural and hydraulic fractures into AI models to predict

productivity improvements, thereby enabling more accurate evaluations of acid fracturing efficiency^[14, 17]. Among these, ANN models have shown superior performance compared to SVM and FL system models. Key parameters influencing acid fracture performance were identified as acid volume, reservoir permeability, and natural fracture spacing^[17].

Statistical methodologies have also been applied in this field, with multiple regression models developed to analyze the impact of parameters on production enhancement. The support vector method emerged as particularly effective for production forecasting^[18]. Adopting a data-driven approach, the gradient boosting decision tree (GBDT) algorithm was utilized to establish a post-acid-fracturing productivity prediction model. This model was trained on geological, engineering, and production testing data from 96 fracturing stages across 15 horizontal wells within the Gaoshiti-Moxi Block in Sichuan Basin. Subsequently, the model was integrated with particle swarm optimization (PSO) to optimize fracturing parameters and predict productivity^[19]. Furthermore, optimization analyses of acid fracturing schemes for carbonate geothermal reservoirs, based on machine learning, have been conducted through the establishment of a coupled thermo-hydro-chemical model. This approach employed the turning band method and Design of Experiments (DOE) sampling, with deep neural network algorithms integrated with multiple optimization techniques to optimize acid fracturing schemes^[20].

In this paper, big data analysis technologies such as Shapley Additive Explanations and Random Forest are utilized to investigate the main controlling factors affecting well production within ultra-deep carbonate reservoirs in the Penglai gas area. An evaluation method based on a comprehensive acid fracturing index is developed, which provides a novel approach to assess the acid fracturing performance of deep carbonate reservoirs and offers theoretical support for optimizing deep acid fracturing designs.

Determinants of Acid Fracturing Performance in Deep Carbonate Reservoirs

Data Set Acquisition and Preprocessing

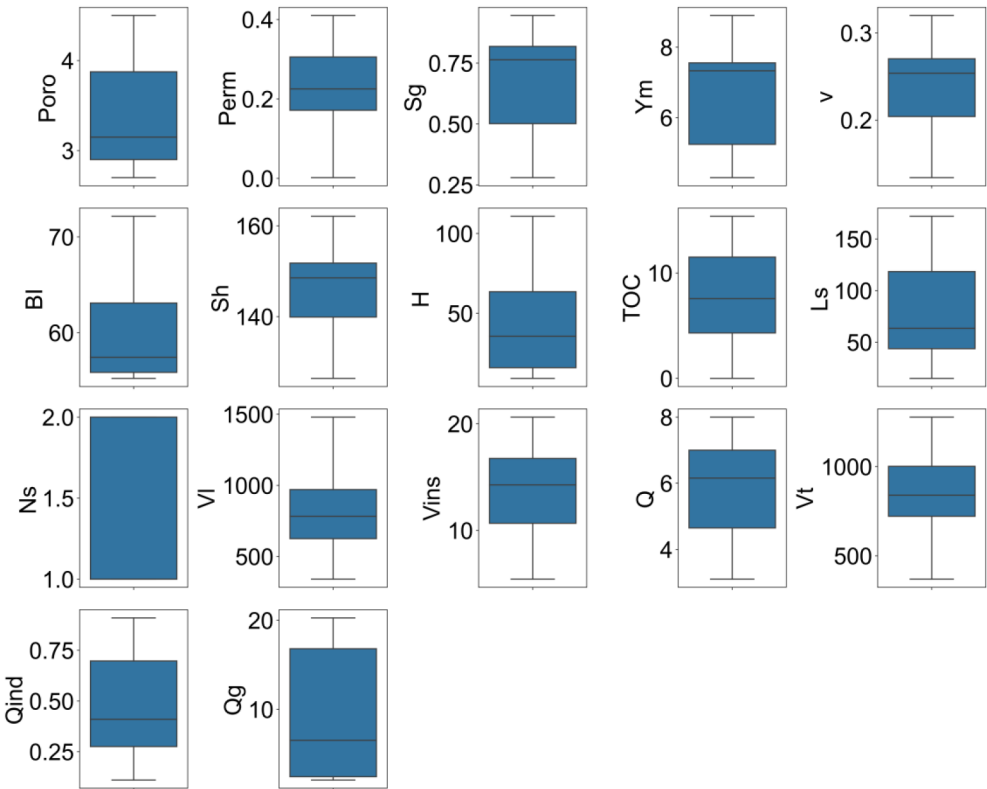
To assess the stimulation efficacy of Deng-4 Formation wells following acid fracturing, this study leverages geological, operational, and flowback data analyzed through a comprehensive data analytics framework. The initial phase involves diagnosing common data integrity challenges—such as gaps, inconsistencies, and inaccuracies—to ensure robustness prior to modeling. Subsequently, data preprocessing is performed to address missing values and outliers, thereby enhancing dataset reliability and strengthening the validity of subsequent data mining outcomes. For a systematic investigation into the critical factors governing treatment success, the Pearson correlation coefficient is employed to quantitatively assess the influence of individual parameters on well performance. This approach allows for the identification of key variables significantly impacting acid fracturing effectiveness in the Deng-4 wells.

A descriptive statistical analysis was performed on key attributes—including well information, geological and construction factors, and test production data—from early acid fracturing wells in the Deng-4 Formation. Metrics such as mean, variance, minimum, maximum, median, and quantiles were calculated to characterize the dataset^[21]. The distribution of these parameters is provided in [Table 1](#).

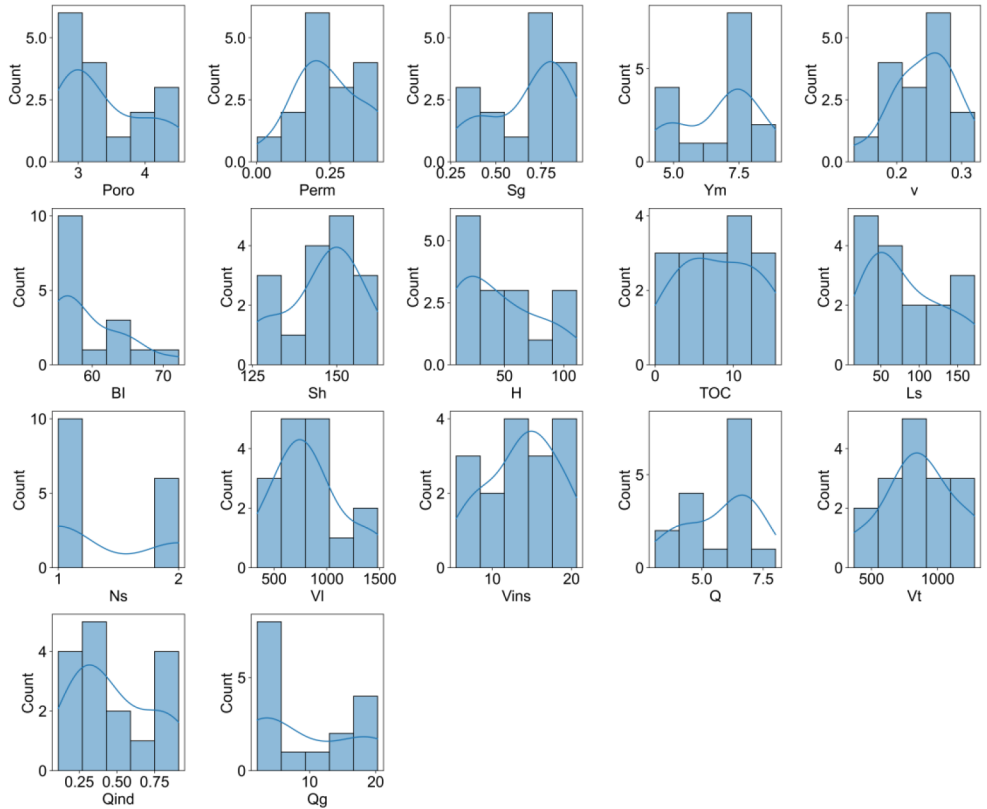
Table 1—Summary of key parameters for the Deng-4 Formation

Parameter	Mean	Variance	Minimum	25% Quantile	50% Quantile	75% Quantile
Poro (Porosity, %)	3.53	1.08	2.40	2.90	3.15	4.03
Perm (Permeability, mD)	0.33	0.20	0.13	0.19	0.25	0.42
Sg (Gas Saturation)	0.75	0.16	0.37	0.74	0.79	0.86
Ym (Young's Modulus, GPa)	6.47	1.83	4.22	4.82	6.29	7.79
ν (Poisson's Ratio)	0.22	0.06	0.13	0.20	0.22	0.26
BI (Brittleness Index)	60.45	7.53	44.69	56.21	61.69	65.03
Sh (Min. Horizontal Stress, MPa)	145.03	12.44	126.40	140.00	144.80	154.20
H (Reservoir Thickness, m)	59.31	25.47	16.80	42.75	55.90	74.45
TOC (Total Organic Carbon, %)	21.06	17.87	0.66	6.82	11.52	35.45
Ns (Number of Stages)	2.27	1.75	1.00	1.00	2.00	3.00
Ls (Treatment Length, m)	106.80	56.48	15.00	65.00	107.00	133.50
Vl (Acid Volume, m ³)	893.67	338.23	340.00	662.50	960.00	1080.00
Vins (Acid Intensity, m ³ /m)	11.97	5.62	5.33	8.42	10.21	14.39
Q (Injection Rate, m ³ /min)	6.01	1.42	3.30	5.40	6.30	7.00
Vt (Total Fluid Volume, m ³)	950.32	349.17	370.00	764.00	989.00	1112.89
Vg (Test Production, 10 ⁴ m ³ /d)	9.06	11.15	0.00	1.83	3.40	13.50

The assessment of feature parameters primarily consists of constant value detection and missing value analysis. A variable possessing only a single value is identified as constant, rendering it ineffective for influencing factor analysis and unsuitable for data mining due to its lack of variability. The evaluation of missing values focuses on the extent of data absence. A high missing rate diminishes a variable's utility for modeling, necessitating data imputation. Specifically, when the proportion of missing values exceeds a certain threshold, the variable contributes little to machine learning modeling and reduces model interpretability. The following strategies are employed for handling missing data: ① When the missing rate is greater than 40% and the attribute importance is low, directly delete the attribute; ② Apply median-based imputation for variables with a low missing rate (below 5%); ③ Use multiple imputation via random forest for variables with a high missing rate and high importance. Data quality determines the effect of the machine learning model, reduces data noise, provides high-quality samples, and ensures the accuracy of machine learning modeling analysis. Processing methods: ① Data points lying beyond the whiskers of a box plot—which mark the dataset's maximum and minimum—are considered outliers; ② The Isolation Forest method recursively and randomly divides the data set until all sample points are isolated, and those with sparse distribution and far from the high-density group are regarded as "outliers". The statistical distribution of the data after final outlier processing is shown in [Figure 1](#).



(a) Box plot



(b) Histogram

Figure 1—Statistical distribution diagram of data after data preprocessing

Nominal value conversion refers to converting features input in string form into numerical data for later data processing. The features to be processed include four types: cementing quality (excellent, qualified, unqualified), reservoir space type, acid type, and acid fracturing process. Their original input data forms are all in string form. Except for a few models such as decision trees that can directly process input in string form, most models must input numerical data to work normally. Here, digital encoding is used, and the encoding correspondence is shown in [Tables 2 to 5](#).

Table 2—Solid Well Quality Nominal Value Code

Cementing quality	Nominal value code
Excellent	0
Qualified	1
Unqualified	2

Table 3—Encoding of storage space type directory values

Reservoir space type	Nominal value code
Pore type	1
Fracture-vug type	2
Medium porosity and high permeability	3

Table 4—Encoding of Acid Type Index Values

Acid type	Nominal value code
Low viscosity acid, high viscosity acid	0
Gelled acid	1
Self-generated acid, gelled acid	2
Self-generated acid, gelled acid, drag reducing acid	3
Diverting acid, gelled acid	4

Table 5—Value Code of Acid Pressing Process

Acid fracturing construction process	Nominal value code
2-stage alternating injection + closed acidizing	0
Two-stage alternating injection	1
Two-stage alternating injection acid fracturing + closed acidizing	2
Alternating injection acid fracturing + temporary plugging	3
Multi-stage alternating injection + closed acidizing	4
Bare tube plug removal acidizing + multi-stage alternating acid fracturing	5
Plug removal acidizing + knot temporary plugging + alternating injection acid fracturing	6
Prepad acid fracturing	7
Prepad acid fracturing + closed acidizing	8
Mechanical layering + bare tube gelled acid acidizing + multi-stage alternating deep acid fracturing	9
Mechanical layering + prepada acid fracturing + closed acidizing	10
Self-generated acid + gelled acid three-stage alternating + closed acidizing process	11

Acid fracturing construction process	Nominal value code
Self-generated acid + gelled acid alternating acid fracturing + temporary plugging process	12
Open hole segmented + 2-stage alternating injection acid fracturing	13
Diverting acid acidizing + temporary plugging	14

Feature Correlation Analysis and Optimization

Prior to developing the machine learning model, multiple exploratory data analysis techniques were applied to perform feature screening on the dataset. Feature collinearity is visualized through a heatmap, as illustrated in [Figure 2](#). Correlation coefficients range from -1 to 1 , with 1 denoting a perfect positive relationship, -1 a perfect negative relationship, and 0 indicating no association. The Pearson correlation coefficient assesses linear relationships between two continuous variables. In contrast, the Spearman correlation coefficient evaluates monotonic relationships without assuming linearity, while the Kendall rank correlation coefficient also measures monotonic association based on concordance among paired observations. Thus, Spearman and Kendall coefficients are particularly suitable when data deviate from normality or contain outliers. The Spearman coefficient captures the strength and direction of monotonic dependence and is moderately robust to outliers. The Kendall coefficient, which considers rank alignment, exhibits even greater robustness to anomalous values. Integrating the properties and applications of these three correlation measures, an averaged correlation matrix is presented in [Figure 2](#). The correlation between the dosage of large particle size and sanding strength, and between the dosage of medium particle size and the dosage of medium viscosity slickwater is greater than 0.7 , indicating a strong correlation. At present, in the fracturing process, medium viscosity slickwater is mainly used to carry medium particle size quartz sand proppant, and large particle size proppant is easy to settle during fracturing, and its dosage has little impact on the supporting effect of fracture conductivity. Therefore, it is chosen to delete the dosage of medium viscosity slickwater and large particle size to remove redundant features, retain the features with high correlation, and the remaining 15 feature parameters form the feature subset.

[Figure 2](#) presents the correlation matrix between post-fracturing gas production and characteristic parameters from the Dengying Formation's fourth member. In correlation analysis, coefficients are conventionally interpreted as follows: $0-0.3$ indicates weak correlation, $0.3-0.6$ moderate correlation, and $0.6-1.0$ strong correlation. Preliminary Pearson correlation analysis revealed the degree of association between each parameter and post-fracturing gas production. Notably, total fluid volume and acid volume exhibit a very strong correlation. To better understand the underlying mechanisms, parameters were categorized into two groups: geological and operational. Geological parameters—such as elastic modulus, Poisson's ratio, and minimum horizontal in-situ stress—reflect reservoir mechanical properties influenced by depositional and tectonic history, directly affecting fracture initiation and propagation. Construction parameters, including acid volume, acid intensity, injection rate, and total fluid volume, are engineering variables that can be optimized operationally.

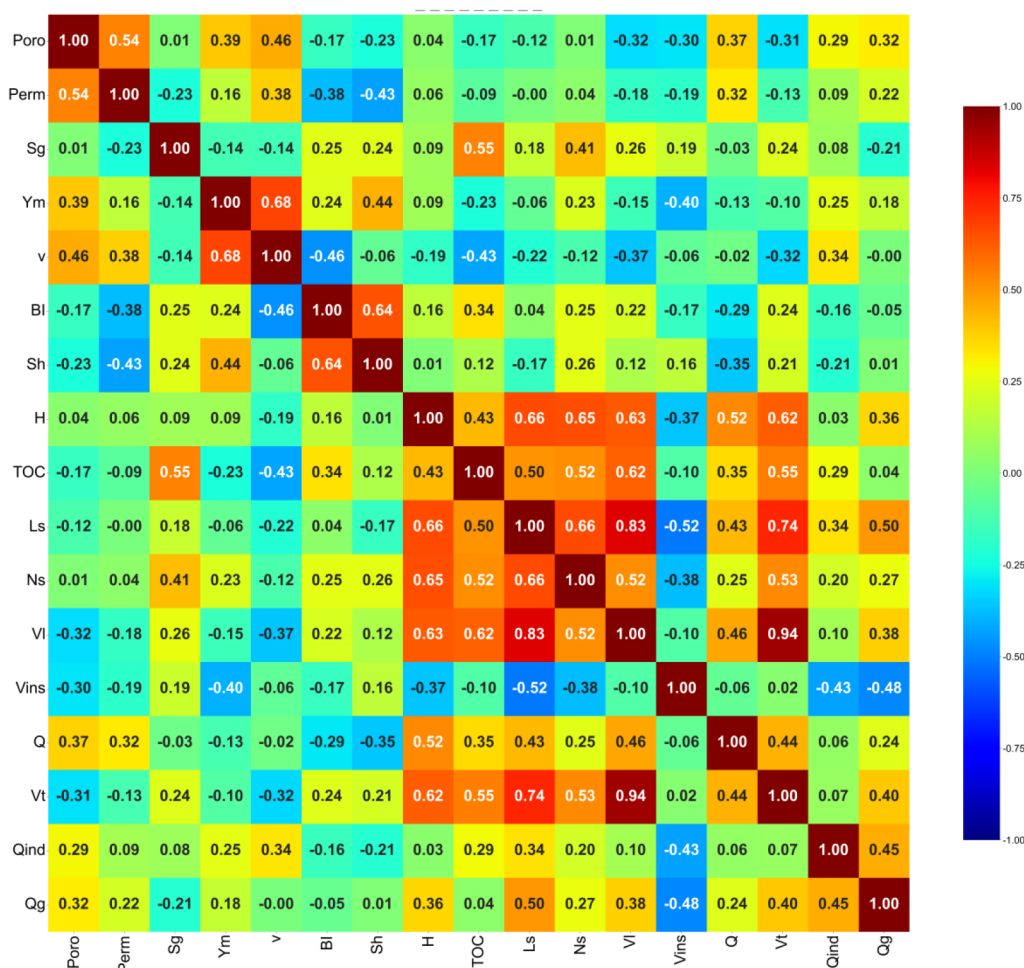


Figure 2—Correlation matrix between post-fracturing test gas volume and characteristic parameters

Correlation visualization among geological/engineering variables and post-stimulation gas production from the Dengying Fm. Fourth Member was performed using the pairplot function in Python's Seaborn library (Figure 3). By drawing a scatter plot matrix between pairwise parameters, the linear and non-linear relationships between each parameter are intuitively presented. The results show that in the construction parameter group, acid volume, acid intensity, displacement and total liquid volume show a significant synergistic effect, and their combined effect has a multiplier effect on the growth of post-fracturing test gas volume; in the geological parameter group, there is a negative coupling relationship between the minimum horizontal in-situ stress and elastic modulus. In a high-stress environment, the reservoir rigidity increases, resulting in limited acid etching. The pairplot maps of the two types of parameters not only quantify the impact of single parameters but also reveal the synergistic mechanism of parameter combinations, providing multi-dimensional data support for building the prediction model of post-fracturing test gas volume of the fourth member of Dengying Formation.

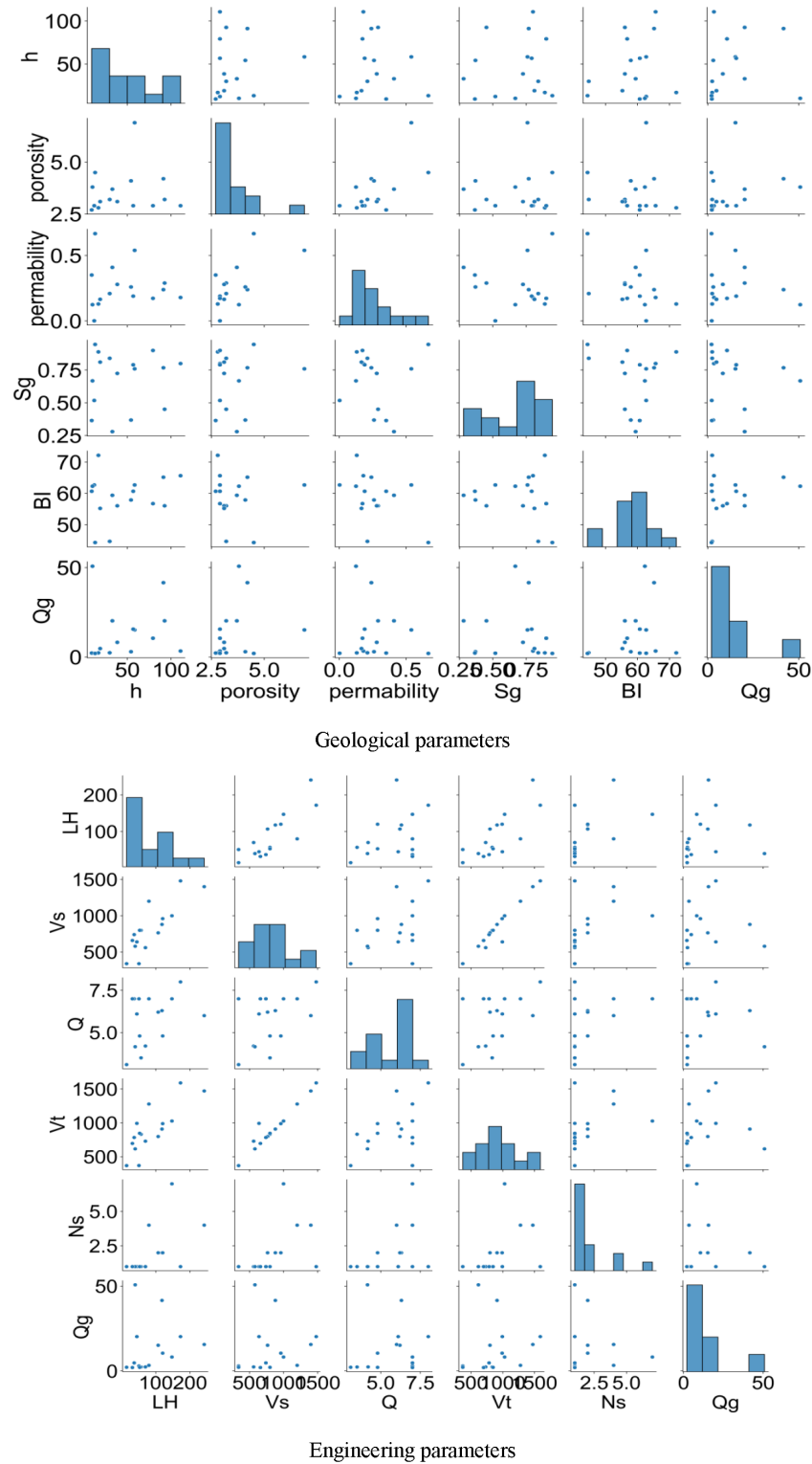


Figure 3—Correlation between post-fracturing gas production and key parameters in the Dy-4 Member

The distribution of characteristic parameters is uneven, and there are large differences in magnitude. To mitigate scaling effects, enhance algorithmic convergence, and boost model precision, the dataset was normalized using Formula 1 prior to training.

$$x = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

In the formula: x is the normalized characteristic parameter data; x_i is the i -th input value; $x_{\min}$ and $x_{\max}$ are the maximum and minimum values of the parameter, respectively.

To mitigate the risk of overfitting due to multicollinearity among feature parameters, the Pearson correlation coefficient was applied to evaluate pairwise feature relationships. Highly correlated features were subsequently removed to enhance the robustness and predictive accuracy of the model for gas content estimation. The calculation of the correlation coefficient follows Formula 2:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (2)$$

In the formula: r represents the correlation coefficient, y_i represents the i -th sample data of the characteristic parameter, $\bar{x}$ and $\bar{y}$ represent the mean value of the characteristic parameter, and n is the number of samples.

The correlation coefficient r is a normalized measure of covariance between two variables, ranging from -1 to 1 . A value of 1 denotes complete positive linear agreement, -1 indicates perfect negative linear agreement, and 0 implies no linear relationship. Various geological and engineering parameters influence fracture volume interpretation, while interactions also exist among these features. To visualize associations between characteristic variables and reconstructed fracture volume, a correlation heatmap is presented in Figure 4. Analysis reveals correlation values ranging from 0.99 to -0.47 , suggesting considerable redundancy within the feature set. Specifically, acid volume, acid intensity, displacement, and total fluid volume exhibit strong interrelationships. To preserve potential explanatory variables while reducing multicollinearity, the actual sand volume was removed, retaining seven geological-engineering parameters. Post-fracturing gas production metrics also show significant correlation among the evaluation indicators.

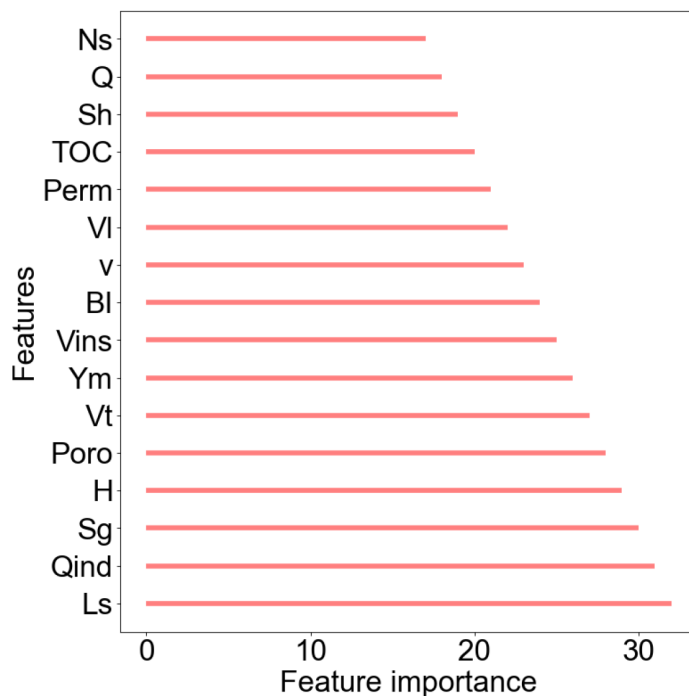


Figure 4—Correlation and absolute value trapezoidal diagram between post-fracturing test production and characteristic parameters

Machine Learning Optimization Model

The performance of machine learning models on unseen data is primarily determined by their generalization capability. Thus, it is essential to incorporate a robust assessment of generalization during model training. A critical procedure involves partitioning the original dataset into training and testing subsets in a rational manner, ensuring that patterns extracted during training remain applicable to new, unseen samples. The test set must be strictly withheld from all stages of model training to prevent overfitting. Generalization performance is commonly measured using prediction accuracy on the test set. In this study, the data were split in an 80:20 ratio, with 80% allocated for model training and the remaining 20% reserved for evaluating generalization.

To evaluate the model's performance in predicting post-fracturing production, multiple evaluation metrics were employed. This study uses mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2) to comprehensively quantify the model's predictive accuracy regarding fracture volume. MSE, RMSE, and MAE gauge prediction error—lower values indicate better accuracy—while R^2 evaluates goodness of fit, with values closer to 1 being desirable. The corresponding formulas are given below:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (3)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (5)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (6)$$

In the formula: n represents the number of coalbed methane content samples; $\hat{y}_i$ represent the predicted value of the i -th sample.

Based on the previously screened geological-engineering characteristic parameters as input variables, regression prediction models of Ridge regression, Decision Tree, Random Forest (RF) and AdaBoost algorithms are constructed respectively to carry out the research on fracture volume inversion prediction and explore the adaptability of different models to the prediction task. Figure 5 illustrates the performance of various predictive models on the test set. Model effectiveness was evaluated using both the fit between predicted and actual values and the coefficient of determination (R^2). An R^2 value closer to 1 indicates stronger predictive accuracy and better model fit. The Random Forest model achieved an R^2 of 0.83, with data points clustering tightly around the identity line, indicating a high quality of fit. By leveraging an ensemble of decision trees combined with random feature selection, the model effectively mitigates overfitting, enhances its ability to learn underlying data patterns, and demonstrates high stability and accuracy in predicting fracture volumes—yielding predictions that align closely with true values across various actual volumes.

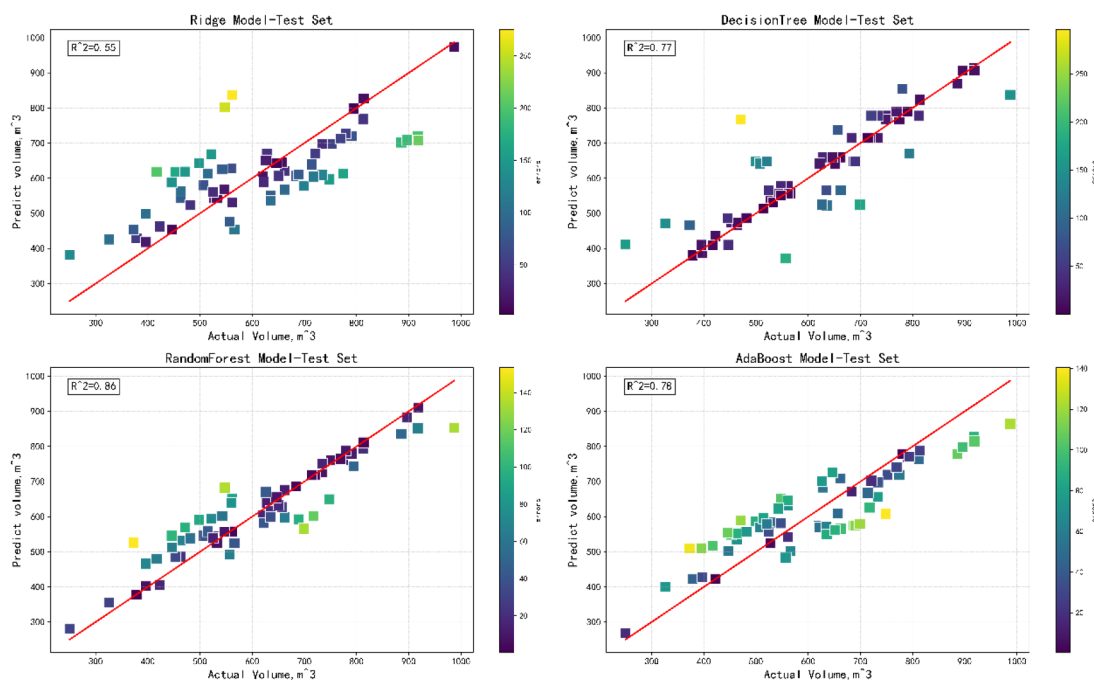


Figure 5—Performance of post-fracturing test production prediction models in the test dataset

Random Forest (RF) is an ensemble learning method based on bagging. Each decision tree can be interpreted as a set of if–then rules, where every path from the root node to a leaf constitutes one rule, with internal nodes defining decision conditions and leaf nodes providing output values. The core idea of RF is to construct multiple decision trees in parallel using bootstrap sampling, while incorporating random feature selection during the training of each tree. Specifically, when splitting nodes, the algorithm randomly selects a subset of features and chooses the best splitting rule from this subset. This approach promotes diversity among the trees, arising not only from random data resampling but also from random feature subspaces. As a result, the differences between individual trees are enhanced, leading to improved generalization in the final ensemble model. Comparative results confirm that the Random Forest regression model attains the highest R^2 on the test set and the best overall fit to the data.

Interpreting and Identifying Main Controlling Factors with SHAP

The SHAP (Shapley Additive Explanations) framework provides a unified approach for interpreting predictions of machine learning models. Rooted in cooperative game theory, it allocates the contribution of each feature to the final prediction similarly to Shapley value computation. SHAP visualization tools not only assess global feature importance but also offer local interpretability by explaining individual predictions, thereby improving model transparency and trustworthiness. The SHAP summary plot (dot plot) displays the distribution of SHAP values per feature. Each horizontal line corresponds to one feature, where each dot represents a sample's SHAP value for that feature. Color indicates the original feature value (high in red, low in blue), and horizontal position reflects the magnitude and direction of the SHAP value. This visualization illustrates both feature importance and the directional effect of the feature on model output. For instance, if high values (red) of a feature cluster on the positive SHAP axis and low values (blue) on the negative axis, this suggests a positive monotonic relationship with the prediction. Conversely, an opposite pattern implies a negative trend. Mixed color distributions may indicate a non-monotonic or complex relationship.

Figure 6 presents the SHAP explanation values and mean absolute SHAP-based importance ranking for the post-fracturing production prediction model. Each point represents one fractured well, and features are sorted vertically by importance. A positive SHAP value denotes a positive contribution to production, while

a negative value indicates a suppressive effect. The driving factors are in turn: treatment section length, acid intensity, acid volume, total hydrocarbon content, total liquid volume, Poisson's ratio, construction displacement, and number of stages. The results show that construction section length, acid intensity, and acid volume are the most important factors affecting the total volume of post-fracturing test production.

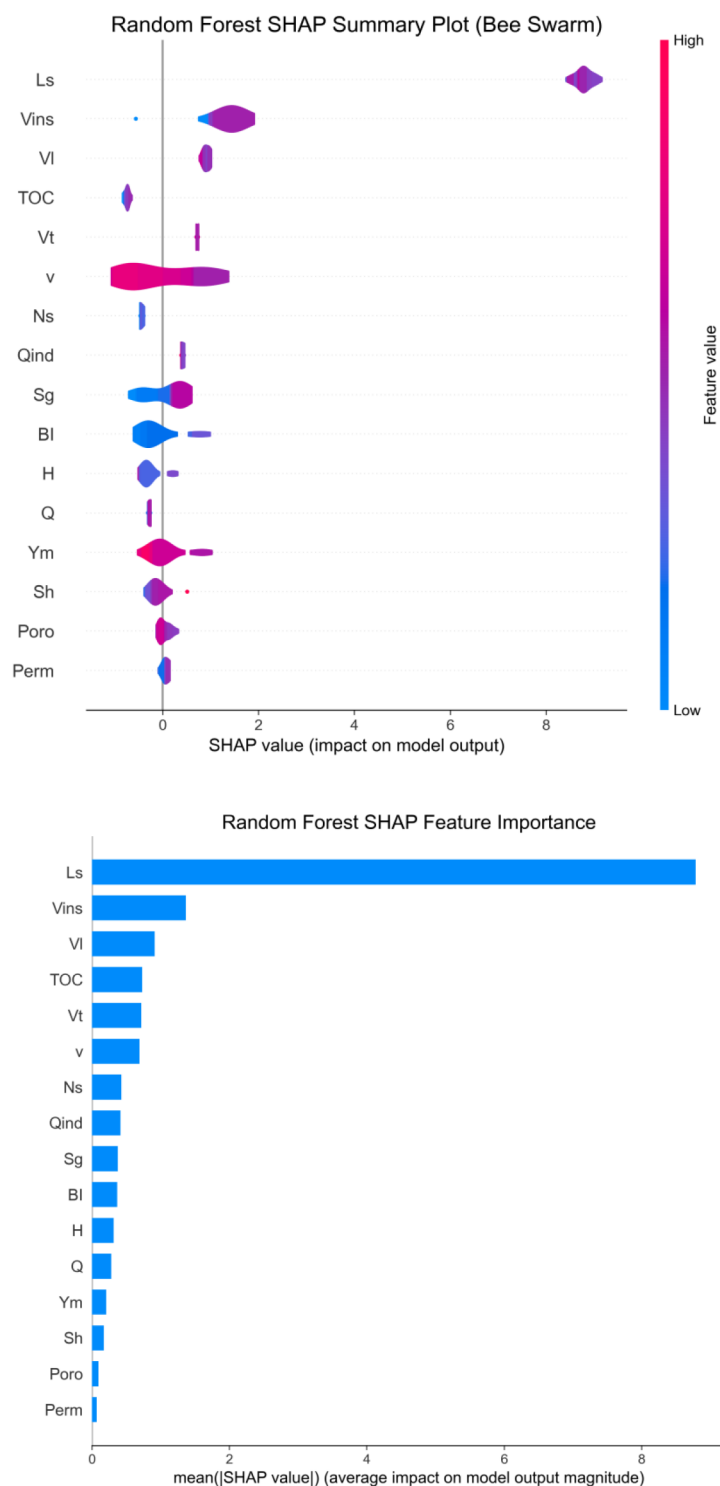


Figure 6—Distribution of SHAP explanation values and ranking of average absolute value importance of the post-fracturing test production model

Comprehensive Evaluation Model of Acid Fracturing Effect Based on AHP-EMW

Based on the identified key factors influencing acid fracturing effectiveness, and integrating multiple post-stimulation evaluation metrics—including water hammer amplitude, decay time, decay rate, pressure drop rate, maximum pressure drop amplitude, number of pressure drops, closure pressure, fracture length, and flowback rate—a comprehensive quantitative evaluation method for acid fracturing effectiveness in deep carbonate reservoirs was developed using a combined Analytic Hierarchy Process (AHP) and Entropy Weight Method (EWM).

AHP Weight Calculation

According to the 1-9 scaling method (Table 6), the influence degree of different parameters on production is judged, and the engineering parameter judgment matrix and geological parameter judgment matrix are established respectively.

Table 6—Meaning of Grade Scale

Scale	Meaning
1	ai is equally important as aj
3	ai is slightly more important than aj
5	ai is much more important than aj
7	ai is more important than aj
9	ai is extremely more important than aj
2,4,6,8	Intermediate values between adjacent judgments
Reciprocal of 1-9	Indicates the importance of aj relative to ai

Compute the principal eigenvalue for judgment matrix Z:

$$Z_{n \times n} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \quad (7)$$

$$\lambda_{\max} = \max\{\lambda \parallel \lambda E - Z \parallel\} \quad (8)$$

Owing to the considerable subjectivity involved in weighting production indicators, it is essential to validate the consistency of the judgment matrix,

$$CI = \frac{\lambda_{\max} - n}{n - 1} \quad (9)$$

$$CR = \frac{CI}{RI} \quad (10)$$

The consistency index (CI), relative consistency index (CR), and average random index (RI) are defined, with their corresponding values provided in Table 7.

Table 7—Average Random Consistency Index (RI) Reference Values

Order n	1	2	3	4	5	6	7	8	9
RI	0	0	0.58	0.89	1.12	1.26	1.36	1.41	1.45

$$CI = \frac{\lambda_{\max} - n}{n - 1} \quad (11)$$

$$CR = \frac{CI}{RI} \quad (12)$$

When $CR < 0.1$, the consistency test is passed.

Solve the maximum eigenvalue corresponding to the judgment matrix:

$$Z\bar{w} = \lambda\bar{w} \quad (13)$$

The weight vector corresponding to each parameter in the judgment matrix is obtained after normalizing its eigenvector:

$$\bar{w}^* = \frac{\bar{w}}{\sum \bar{w}} \quad (14)$$

EWM Weight Calculation

Entropy is a method to measure the effective information provided by the system. It matches the method by which decision-makers make decisions by obtaining data to reflect the disorder degree of the system. An index with lower entropy supplies more substantial information and is assigned a higher weight; conversely, greater entropy corresponds to diminished informational utility and a reduced weight. Consequently, the entropy weight method excels at extracting underlying patterns from datasets, mitigates subjective bias, and enhances the objectivity and reliability of evaluation outcomes. In summary, this method offers a data-driven approach to assign index weights, thereby improving the robustness and credibility of the evaluation.

Due to the different units and dimensions of each parameter, as well as the differences in their value sizes and effective ranges, it is necessary to normalize these parameters first.

Calculation formula for positive indicators:

$$S = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (15)$$

Calculation formula for negative indicators:

$$S = \frac{X_{\max} - X}{X_{\max} - X_{\min}} \quad (16)$$

In the formula, S denotes the normalized value of the parameter; represent the maximum and minimum values of the parameter across the study area, respectively, while X corresponds to the parameter value within the target interval.

Index proportion calculation: convert the original data with m indexes and n records into an $n \times m$ matrix $R = (r_{ij})_{n \times m}$, and calculate the proportion p_{ij} of the i -th row record in the j -th index in matrix R .

$$p_{ij} = \frac{r_{ij}}{\sum_{i=1}^n r_{ij}} \quad (17)$$

For a problem containing m indexes and n records, the information entropy h_j of the j -th index is expressed as:

$$h_j = \frac{\sum_{i=1}^n p_{ij} \ln p_{ij}}{\ln n} \quad (18)$$

It is stipulated that when $p_{ij}=0$, $p_{ij} \ln p_{ij}=0$, so $h_j \in [0, 1]$.

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After obtaining the entropy value corresponding to each index, calculate the weight of each index as follows:

$$d_j = 1 - h_j \quad (19)$$

$$w_i = 1 - \frac{h_j}{\sum_{j=1}^m d_j} \quad (20)$$

AHP-EWM Weight Calculation

Combining AHP and EWM, the importance of each parameter can be judged through subjective experience, and the important information of each parameter can be objectively reflected, making the calculated weight more scientific. The comprehensive weight is calculated by the following formula:

$$\lambda_i = \frac{\bar{w}^* w_i}{\sum_{i=1}^n \bar{w}^* w_i} \quad (21)$$

The weight coefficients of each method are calculated according to the calculation steps of AHP and EWM, as shown in Table 8. It is found that among the quantitative parameters of post-acid fracturing construction data and flowback production data, the type of well test curve, number of pressure drops, maximum pressure drop amplitude, and fracture length have a relatively greater impact on the effect.

Table 8—Weighting Coefficients of Acidizing Effectiveness using AHP, EWM, and AHP-EWM Methods

Method	Weight								
	Amplitude	Decay time	Decay rate	Fracture length	Pressure drop rate	Maximum pressure drop amplitude	Number of pressure drops	Closure pressure	Flowback rate
AHP	0.06	0.06	0.06	0.18	0.11	0.11	0.11	0.12	0.19
EWM	0.06	0.1	0.2	0.07	0.11	0.14	0.2	0.06	0.05
AHP-EWM	0.04	0.06	0.12	0.13	0.12	0.15	0.22	0.07	0.09

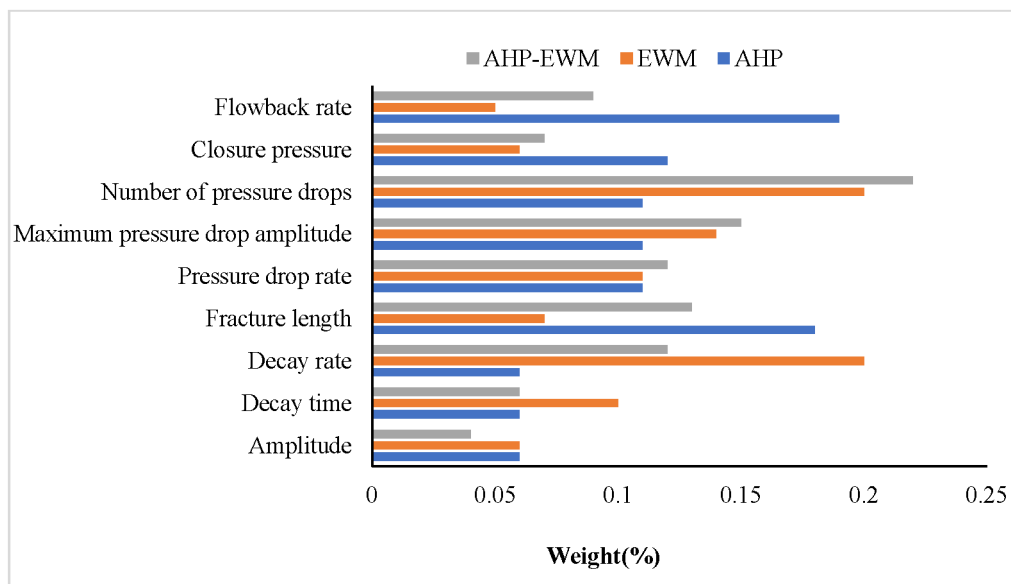


Figure 7—Acid fracturing effect AHP, EWM, AHP-EWM method weight

Acid fracturing effect index model:

$$AI=(0.04 \ 0.06 \ 0.12 \ 0.13 \ 0.12 \ 0.15 \ 0.22 \ 0.07 \ 0.09) \begin{pmatrix} P_a \\ t \\ v_p \\ L \\ v \\ P \\ N \\ P_c \\ B_r \\ W_t \end{pmatrix} \tag{22}$$

Where: P_a , t , v_p , L , v , P , N , P_c , B_r , W_t are the normalized water hammer amplitude, decay time, decay rate, pressure drop rate, maximum pressure drop amplitude, number of pressure drops, closure pressure, fracture length, and flowback rate, respectively.

Table 9—Weighting Coefficients of Geological Factors using AHP, EWM, and AHP-EWM Methods

Method	Weight					
	Porosity	Permeability	Gas Saturation	Reservoir Thickness	Reservoir Quality Index	Total Hydrocarbon Content
AHP	0.22	0.09	0.09	0.21	0.2	0.2
EWM	0.12	0.12	0.12	0.36	0.22	0.06
AHP-EWM	0.15	0.06	0.06	0.42	0.24	0.07

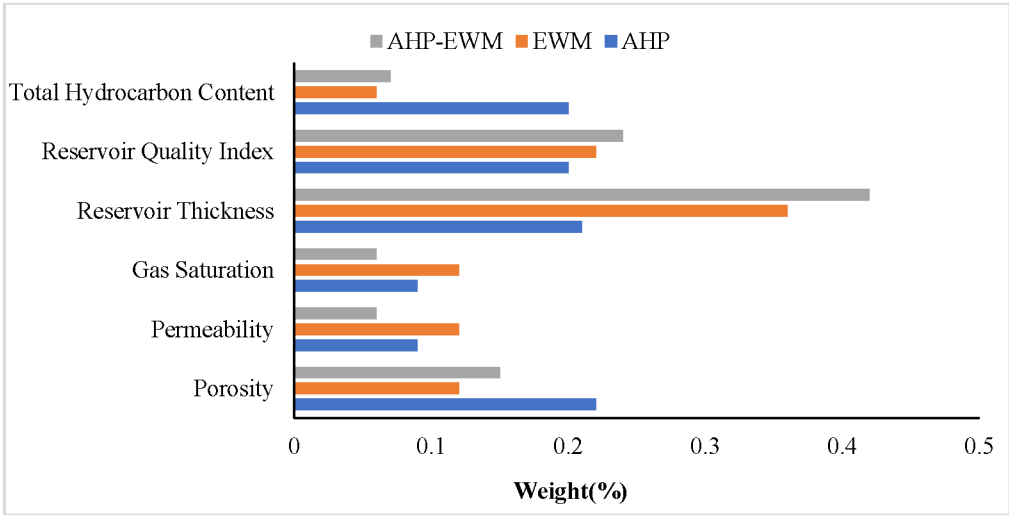


Figure 8—Geological AHP, EWM, AHP-EWM method weight coefficient

Geological index model:

$$FI=(0.15 \ 0.06 \ 0.06 \ 0.42 \ 0.24 \ 0.07) \begin{pmatrix} \varphi \\ K \\ S_g \\ H \\ N \\ TPH \end{pmatrix} \tag{23}$$

Where: φ , K , S_g , H , N , are the normalized porosity, permeability, gas saturation, reservoir thickness, reservoir quality index, and total hydrocarbon content, respectively.

Finally, by combining the acid fracturing effect index and the geological index, the comprehensive index is calculated as the average to characterize the acid fracturing effect of each well in the Penglai gas field.

$$\text{composite index} = 1/2 \left[\begin{matrix} 0.14 \\ 0.06 \\ 0.12 \\ 0.13 \\ 0.12 \\ 0.15 \\ 0.22 \\ 0.07 \\ 0.09 \end{matrix} \right]^T \begin{matrix} P_a \\ t \\ v_p \\ L \\ v \\ p \\ N \\ P_c \\ B_r \end{matrix} + \begin{matrix} 0.15 \\ 0.06 \\ 0.06 \\ 0.42 \\ 0.24 \\ 0.07 \end{matrix} \begin{matrix} \varphi \\ K \\ S_g \\ H \\ M \\ TPH \end{matrix} \right] \quad (24)$$

Case Study

The carbonate reservoirs within the PL gas field are situated at ultra-deep depths ranging from 6000 to 8000 meters, categorizing them as ultra-deep gas accumulations. With formation temperatures between 160 °C and 180 °C, these reservoirs qualify as high-temperature systems. The formation pressure coefficient, varying from 1.03 to 1.06, indicates normal pressure conditions. These reservoirs exhibit low porosity (2%–6%) and permeability ranging from low to ultra-low (0.007–9.8 mD). Primary storage spaces consist of pores and vugs, with fractures developed locally. Heterogeneous vertical stress profiles across individual wells pose significant challenges to achieving uniform stimulation along the wellbore. Using 13 pre-acid-fracturing wells in the Penglai gas field as a case study, a comprehensive evaluation integrating both engineering effectiveness indices and geological parameters was conducted to assess the post-fracturing production performance, as summarized in Table 10.

Table 10—Acidizing Effect Index and Test Production for Wells in Penglai Gas Area

Well Name	Acid Fracturing Effect Index	Geological Index	Comprehensive Index	Test Production
Db 1 Upper	0.47	0.55	0.51	20.28
Db 1 Lower	0.45	0.25	0.35	8.26
PS 1	0.28	0.30	0.29	3.4
PS 2	0.23	0.15	0.19	0
PS 6	0.25	0.25	0.25	2.4
PS 7	0.28	0.20	0.24	2.4
PS 8	0.91	0.37	0.64	41.6
PS 9	0.28	0.42	0.35	15.22
PS 10	0.35	0.43	0.39	11.79
PS 12	0.71	0.11	0.41	15.66
PS 13	0.22	0.46	0.34	10.56
PS 1X5	0.48	0.14	0.31	2.45
PS 1X6	0.31	0.29	0.30	3.03

Linear relationships between the acid fracturing effect index, geological index, comprehensive index, and test production of each well in the Penglai gas field were established, as shown in Figure 11. It is found that the correlation between the geological index and test production is the weakest, while the correlation between the acid fracturing effect index and test production is relatively stronger. However, the comprehensive index obtained by combining the geological index and the acid fracturing effect index shows an extremely strong relationship with test production, indicating that the test production level is the

combined result of engineering effects and geological foundations. Both effective treatment design and quality field operation are vital to ensuring stimulation success in the Penglai gas field. The high consistency ($> 90\%$) between the comprehensive index from our quantitative model and the production data proves its robustness and establishes it as a reliable tool for evaluating acid fracturing outcomes in the area's carbonate reservoirs.

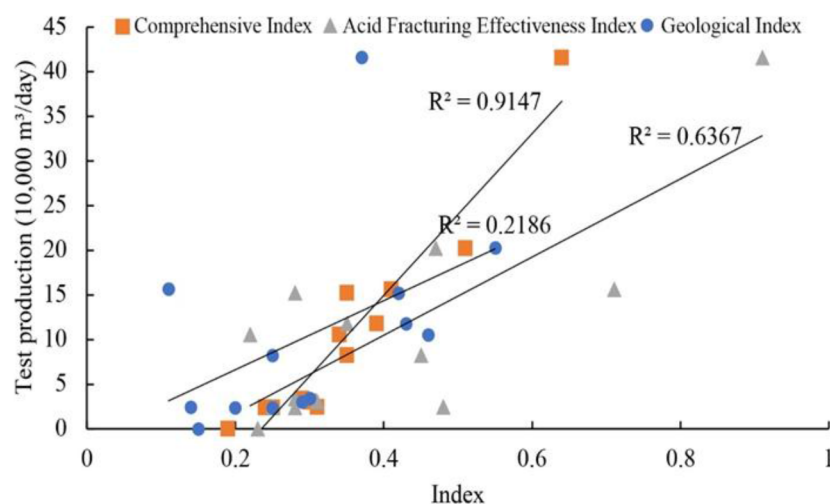


Figure 9—Linear Regression of Comprehensive Acidizing Effect Index and Test Production in Lamp IV Section

Conclusions

In this paper, a data-driven approach for post-acid fracturing evaluation in deep carbonate formations is proposed. Leveraging the geological, construction, and production data of the pre acid-fracturing wells in the PL gas block, data preprocessing is carried out to improve data quality. Among ridge regression, Decision Tree, Random Forest and AdaBoost algorithms, Random Forest regression model obtains the highest predictive accuracy, then, the SHAP (Shapley Additive Explanations) model is used to determine the importance ranking of well area parameters, The outcomes are integrated via weighted averaging to establish the significance order of reservoir and treatment parameters, thereby informing the selection of dominant controlling factors for neural network modeling. To further analyze the key factors affecting acid fracturing treatment outcomes, establishes a multivariate quantitative evaluation model for acid fracturing effects by integrating Analytic Hierarchy Process and Entropy Weight Method, clarifies the main controlling factors affecting acid fracturing effects, and forms a new quantitative evaluation method for carbonate acid fracturing wells in PL gas area. Results demonstrate that treatment section length, acid intensity, acid volume, total hydrocarbon content, pressure drop frequency, significantly impact fracturing effectiveness. The constructed quantitative acid fracturing effectiveness index achieved over 90% linear correlation with actual well test production, verifying model validity.

This article presents a novel methodology for post evaluation of acid fracturing performance in deep and ultra-deep carbonate reservoirs, and provides a theoretical basis for efficient deep acid fracturing design.

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References

1. Guo, J. C., Guan, C. C., Li, X., et al The core concept and key technologies of three-dimensional acid fracturing for deep sulfur-bearing carbonate reservoirs in the Sichuan Basin[J]. *Natural Gas Industry*, 2023,**43**(9), 14–24.
2. Yang, Y., Wen, L., Xie, J. R., et al Progress and directions in marine carbonate gas exploration in the Sichuan Basin[J]. *China Petroleum Exploration*, 2020,**25**(3), 44–55.
3. Ma, X. H., Yang, Y., Wen, L., et al Distribution and exploration direction of medium- and large-sized marine carbonate gas fields in Sichuan Basin[J]. *Petroleum Exploration and Development*, 2019,**46**(1), 1–13.
4. Zhang, Y., & Wu, W. Q. Application status of acid fracturing technology in deep carbonate rocks[J]. *Chemical Engineering & Equipment*, 2023,(2), 53–54.
5. Zhu T, Wei X, Zhang Z. Numerical simulation of hydraulic-mechanical-chemical field coupled acid fracturing in complex carbonate reservoir[J]. *Computers and geotechnics*, 2023.
6. Ueda K, Zhang W, Zhu D, et al Evaluation of Acid Fracturing by Integrated Pressure Analysis and 3D Simulation: A Field Application for Multi-Stage Stimulation in Horizontal Wells: SPE Hydraulic Fracturing Technology Conference[C], 2016-02-09.
7. Xu L J, Zhang S C, Mou J Y. Acid Leakoff Mechanism in Acid Fracturing of Naturally Fractured Carbonate Gas Reservoirs[J]. *Advanced Materials Research*, 2014,**868**:682–685.
8. Cao X, Ren J, Xin S, et al Study of Acid Fracturing Strategy with Integrated Modeling in Naturally Fractured Carbonate Reservoirs[J]. *Processes*, 2024,**12**(4).
9. Wang T, Yang Y, Peng Y, et al Simulation and evaluation for acid fracturing of carbonate reservoirs based on embedded discrete fracture model[J]. *Natural Gas Industry B*, 2021,**8**(6):637–649.
10. Xiong D, Zhang L, Xiao C, et al A novel post-fracturing evaluation method for fractured-vuggy carbonate reservoir by using pumping-stop pressure drop[J]. *Geoenergy Science and Engineering*, 2025,**254**:214037.
11. Desouky M, Aljawad M.S, Alhoori H, et al Evaluating the Effects of Acid Fracture Etching Patterns on Conductivity Estimation Using Machine Learning Techniques: SPE Europec[C], 2020-12-01.
12. Eliebid M, Hassan A M, Mahmoud M, et al Intelligent Prediction of Acid-fracturing Performance in Carbonates Reservoirs: SPE Kingdom of Saudi Arabia Annual Technical Symposium and Exhibition[C], 2018-04-23.
13. Hassan A M, Aljawad M S, Mahmoud M A. Predicting the Productivity Enhancement After Applying Acid Fracturing Treatments in Naturally Fractured Reservoirs Utilizing Artificial Neural Network: Abu Dhabi International Petroleum Exhibition & Conference[C], 2021-11-15.
14. Hassan A, Aljawad M S, Mahmoud M. An Artificial Intelligence-Based Model for Performance Prediction of Acid Fracturing in Naturally Fractured Reservoirs[J]. *ACS Omega*, 2021,**6**(21):13654–13670.
15. Al-Sulaiman M, Aljawad M S, Hassan A, et al Machine Learning Prediction of Acid Fracture Performance in Naturally Fractured Dolomite Formations[J]. *Arabian Journal for Science and Engineering*, 2023,**48**(12):16805–16818.
16. Wang S, Chen Z, Chen S. Applicability of deep neural networks on production forecasting in Bakken shale reservoirs[J]. *Journal of Petroleum Science and Engineering*, 2019, **179**: 112–125.
17. Al-Sulaiman M, Aljawad M S, Hassan A, et al Machine Learning Prediction of Acid Fracture Performance in Naturally Fractured Dolomite Formations[J]. *Arabian Journal for Science and Engineering*, 2023,**48**(12):16805–16818.

18. Kochnev A, Torsunova D V, Boyarov V, et al Possibility of using machine learning algorithms to assess the effectiveness of acid hydraulic fracturing[C], 2022-01-01.
19. Yang Jian, Wang Qian, Zhou Wengao, et al Evaluating staged acid fracturing of horizontal wells in carbonate gas reservoirs based on GBDT algorithm[J]. *Natural Gas Exploration and Development*, 2023, **46**(3): 84–91.
20. Liu Jia, Li Weihua, Xue Yi, et al Optimization Analysis of Acid Fracturing Scheme for Carbonatite Geothermal Reservoir Based on Machine Learning[J]. *Advanced Engineering Sciences*, DOI: [10.12454/j.jsuese.202400761](https://doi.org/10.12454/j.jsuese.202400761).
21. Zhong G H, Xie B, Yang X B, et al Research on logging evaluation methods for shale gas in XX area of Sichuan Basin[C]. Proceedings of 2013 National Natural Gas Academic Annual Conference. 2013: 267–272.